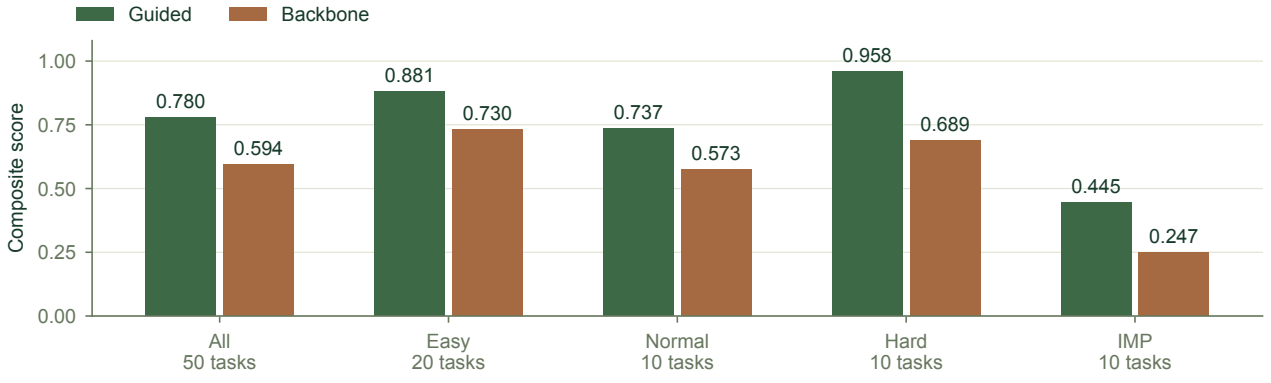
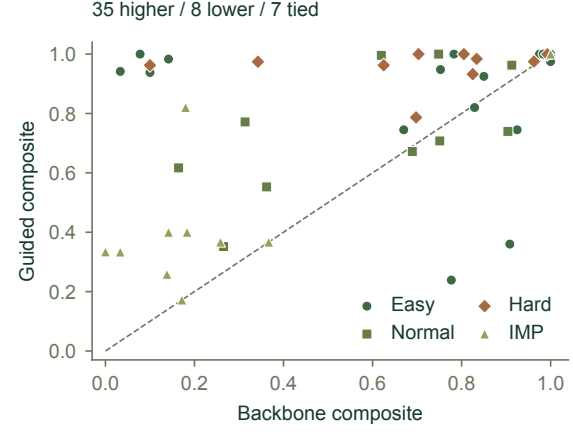
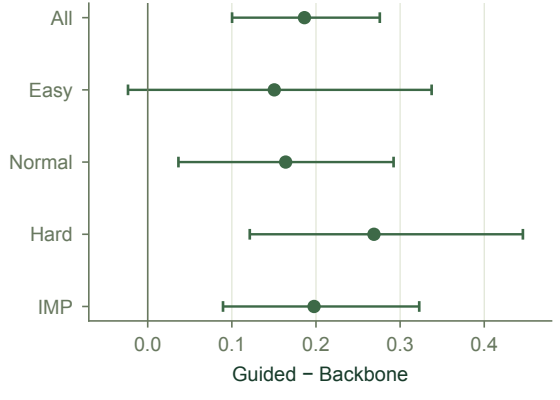
a **Matched backbone comparison**

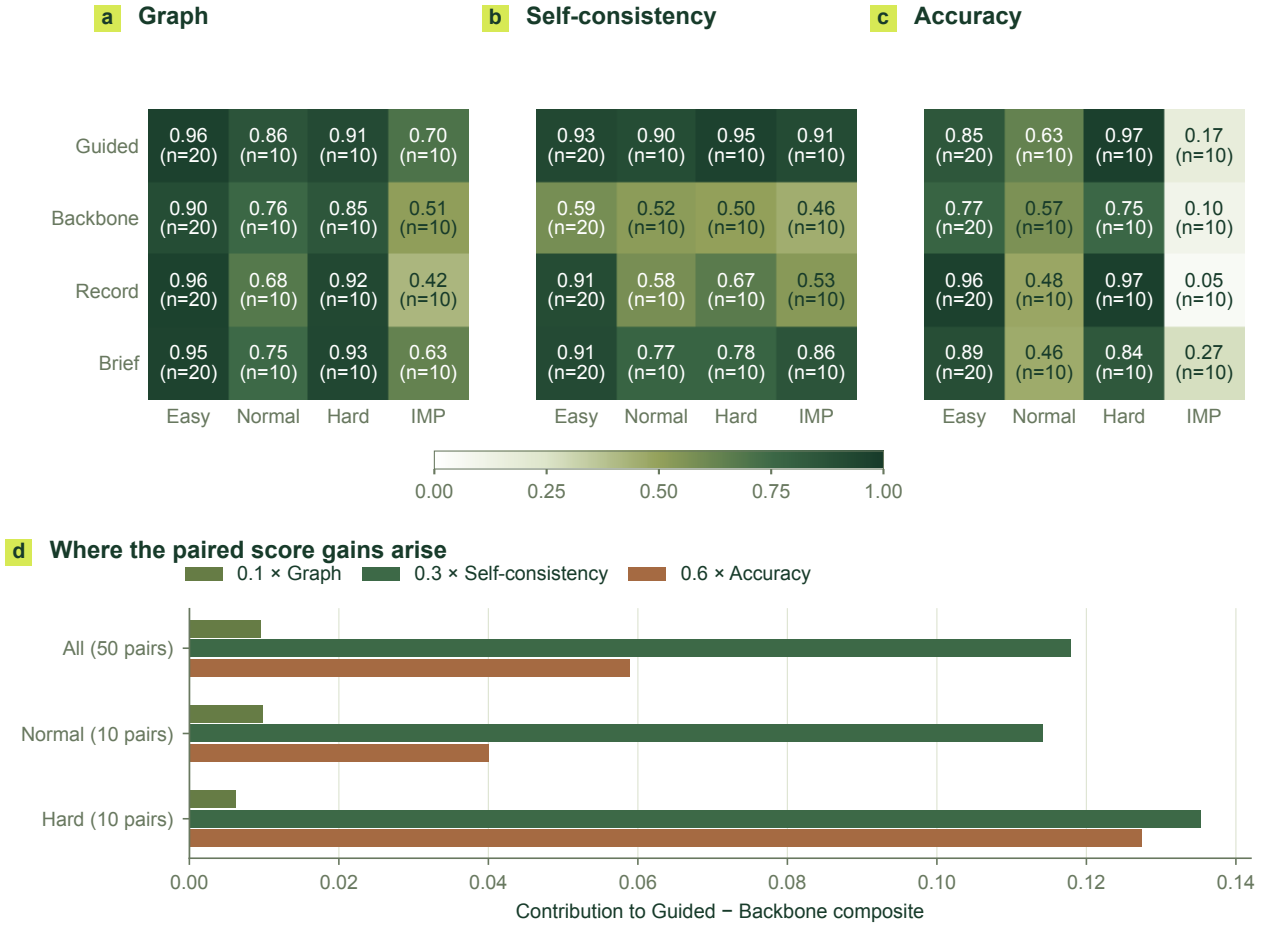


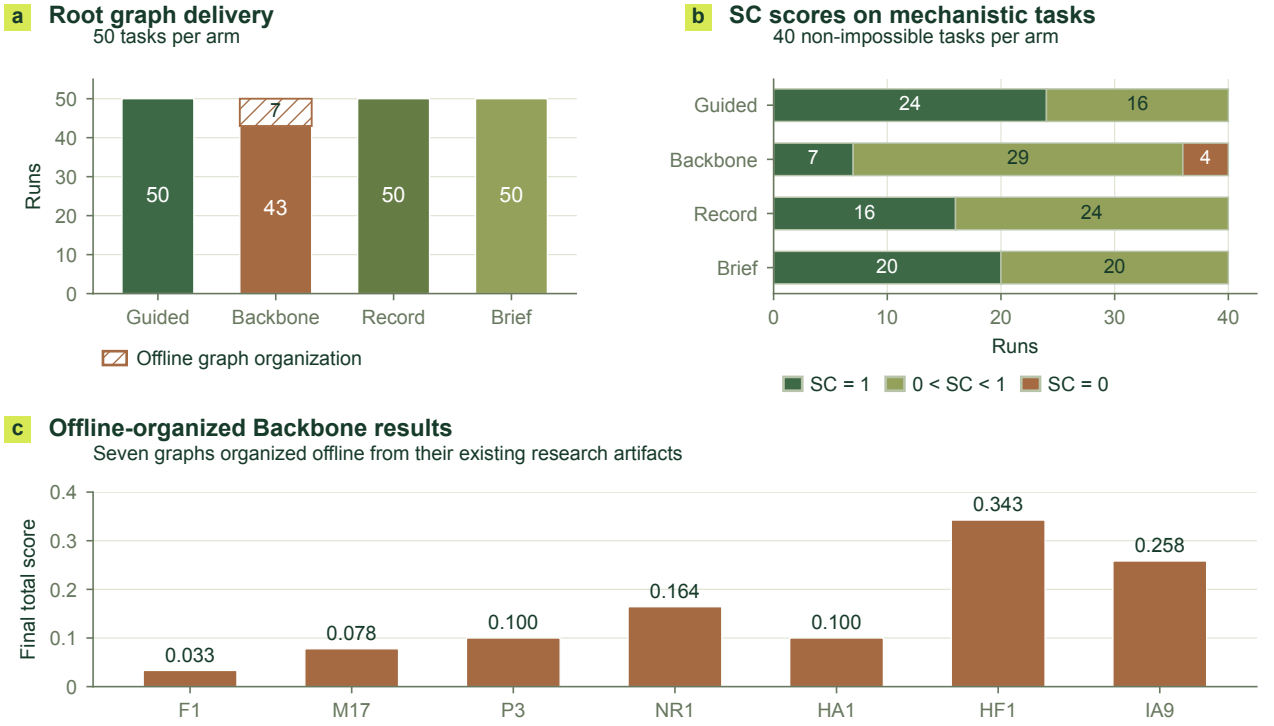
b **Every task retained**

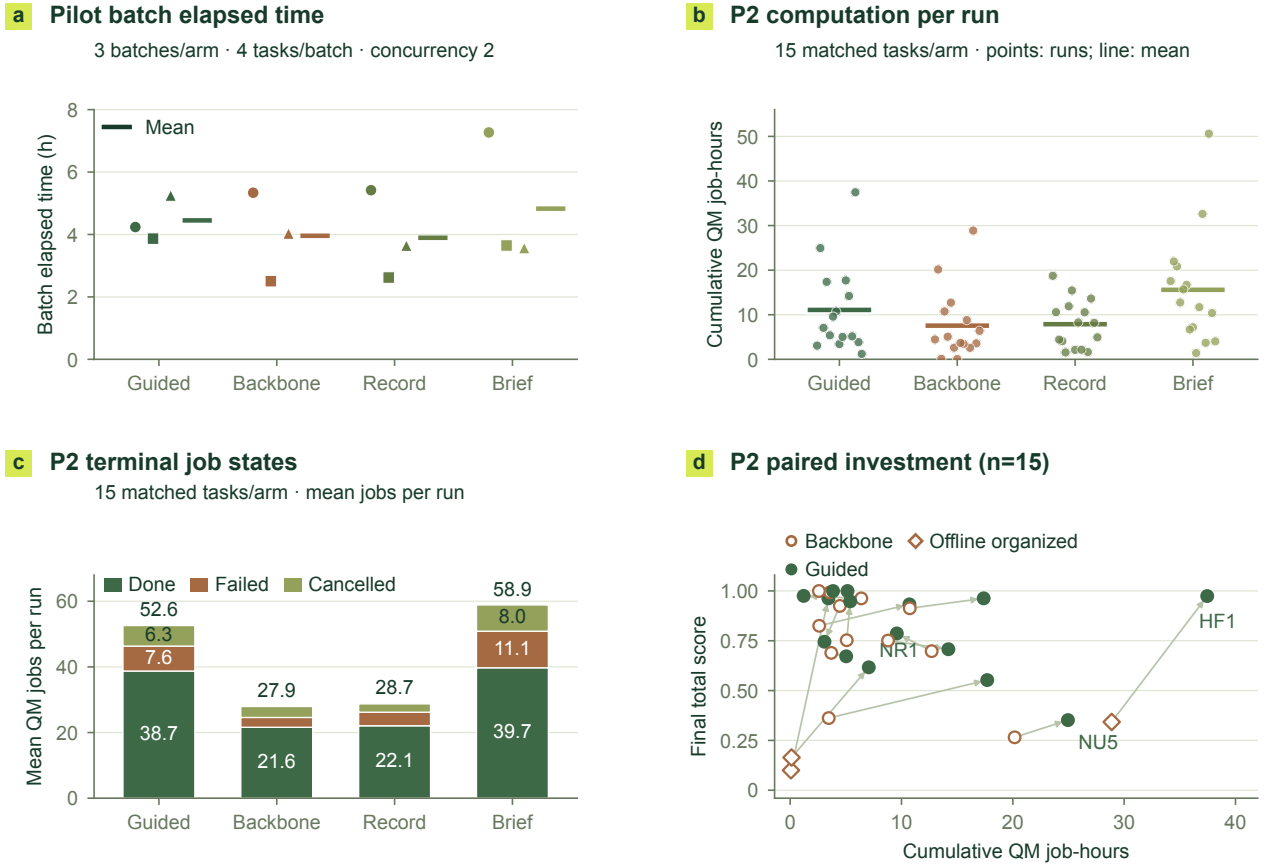


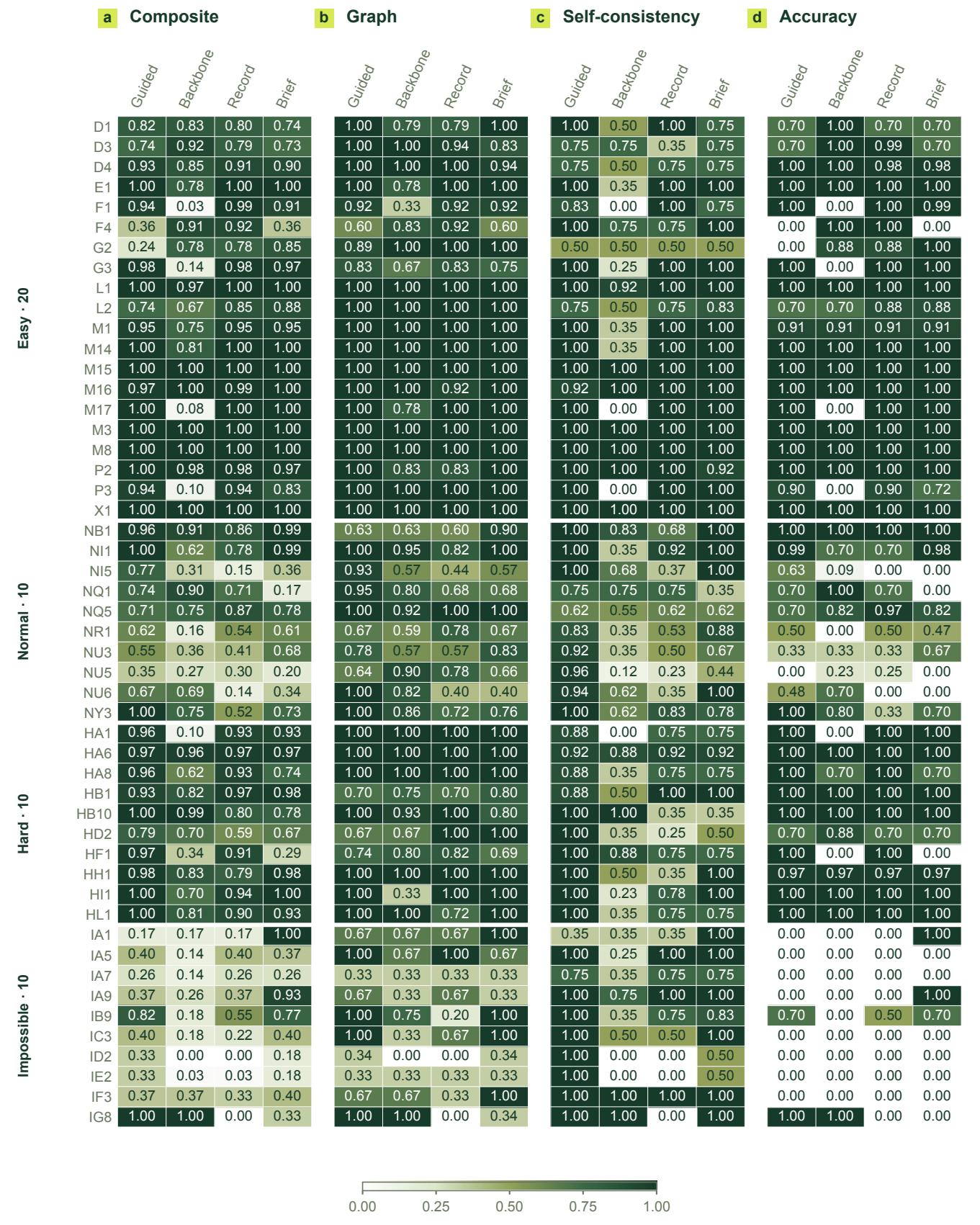
c **Paired gains and 95% intervals**



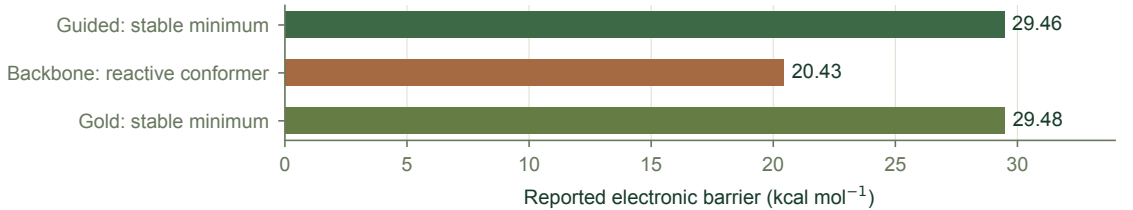

a Graph
b Self-consistency
c Accuracy
Guided
Backbone
Record
Brief
0.96 (n=20)
0.86 (n=10)
0.91 (n=10)
0.70 (n=10)
0.90 (n=20)
0.76 (n=10)
0.85 (n=10)
0.51 (n=10)
0.96 (n=20)
0.68 (n=10)
0.92 (n=10)
0.42 (n=10)
0.95 (n=20)
0.75 (n=10)
0.93 (n=10)
0.63 (n=10)
0.93 (n=20)
0.90 (n=10)
0.95 (n=10)
0.91 (n=10)
0.59 (n=20)
0.52 (n=10)
0.50 (n=10)
0.46 (n=10)
0.91 (n=20)
0.58 (n=10)
0.67 (n=10)
0.53 (n=10)
0.91 (n=20)
0.77 (n=10)
0.78 (n=10)
0.86 (n=10)
0.85 (n=20)
0.63 (n=10)
0.97 (n=10)
0.17 (n=10)
0.77 (n=20)
0.57 (n=10)
0.75 (n=10)
0.10 (n=10)
0.96 (n=20)
0.48 (n=10)
0.97 (n=10)
0.05 (n=10)
0.89 (n=20)
0.46 (n=10)
0.84 (n=10)
0.27 (n=10)
Easy
Normal
Hard
IMP
0.00
0.25
0.50
0.75
1.00
d Where the paired score gains arise
0.1 × Graph
0.3 × Self-consistency
0.6 × Accuracy
All (50 pairs)
Normal (10 pairs)
Hard (10 pairs)
0.00
0.02
0.04
0.06
0.08
0.10
0.12
0.14
Contribution to Guided − Backbone composite


a
Root graph delivery
50 tasks per arm
Runs
50
43
7
50
50
Guided
Backbone
Record
Brief
Offline graph organization
b
SC scores on mechanistic tasks
40 non-impossible tasks per arm
Guided
Backbone
Record
Brief
24
16
7
29
4
16
24
20
20
Runs
SC = 1
0 < SC < 1
SC = 0
c
Offline-organized Backbone results
Seven graphs organized offline from their existing research artifacts
Final total score
0.033
0.078
0.100
0.164
0.100
0.343
0.258
F1
M17
P3
NR1
HA1
HF1
IA9


a
Pilot batch elapsed time
3 batches/arm · 4 tasks/batch · concurrency 2
Batch elapsed time (h)
Mean
Guided
Backbone
Record
Brief
b
P2 computation per run
15 matched tasks/arm · points: runs; line: mean
Cumulative QM job-hours
c
P2 terminal job states
15 matched tasks/arm · mean jobs per run
Mean QM jobs per run
Done
Failed
Cancelled
52.6
6.3
7.6
38.7
27.9
21.6
28.7
22.1
58.9
8.0
11.1
39.7
d
P2 paired investment (n=15)
Backbone
Offline organized
Guided
Final total score
NR1
HF1
NU5
Cumulative QM job-hours


a Composite
b Graph
c Self-consistency
d Accuracy
Guided
Backbone
Record
Brief
Easy · 20
Normal · 10
Hard · 10
Impossible · 10
D1 0.82 0.83 0.80 0.74 1.00 0.79 0.79 1.00 1.00 0.50 1.00 0.75 0.70 1.00 0.70 0.70
D3 0.74 0.92 0.79 0.73 1.00 1.00 0.94 0.83 0.75 0.75 0.35 0.75 0.70 1.00 0.99 0.70
D4 0.93 0.85 0.91 0.90 1.00 1.00 1.00 0.94 0.75 0.50 0.75 0.75 1.00 1.00 0.98 0.98
E1 1.00 0.78 1.00 1.00 1.00 0.78 1.00 1.00 1.00 0.35 1.00 1.00 1.00 1.00 1.00 1.00
F1 0.94 0.03 0.99 0.91 0.92 0.33 0.92 0.92 0.83 0.00 1.00 0.75 1.00 0.00 1.00 0.99
F4 0.36 0.91 0.92 0.36 0.60 0.83 0.92 0.60 1.00 0.75 0.75 1.00 0.00 1.00 1.00 0.00
G2 0.24 0.78 0.78 0.85 0.89 1.00 1.00 1.00 0.50 0.50 0.50 0.50 0.00 0.88 0.88 1.00
G3 0.98 0.14 0.98 0.97 0.83 0.67 0.83 0.75 1.00 0.25 1.00 1.00 1.00 0.00 1.00 1.00
L1 1.00 0.97 1.00 1.00 1.00 1.00 1.00 1.00 1.00 0.92 1.00 1.00 1.00 1.00 1.00 1.00
L2 0.74 0.67 0.85 0.88 1.00 1.00 1.00 1.00 0.75 0.50 0.75 0.83 0.70 0.70 0.88 0.88
M1 0.95 0.75 0.95 0.95 1.00 1.00 1.00 1.00 1.00 0.35 1.00 1.00 0.91 0.91 0.91 0.91
M14 1.00 0.81 1.00 1.00 1.00 1.00 1.00 1.00 1.00 0.35 1.00 1.00 1.00 1.00 1.00 1.00
M15 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00
M16 0.97 1.00 0.99 1.00 1.00 1.00 0.92 1.00 0.92 1.00 1.00 1.00 1.00 1.00 1.00 1.00
M17 1.00 0.08 1.00 1.00 1.00 0.78 1.00 1.00 1.00 0.00 1.00 1.00 1.00 0.00 1.00 1.00
M3 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00
M8 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00
P2 1.00 0.98 0.98 0.97 1.00 0.83 0.83 1.00 1.00 1.00 1.00 0.92 1.00 1.00 1.00 1.00
P3 0.94 0.10 0.94 0.83 1.00 1.00 1.00 1.00 1.00 0.00 1.00 1.00 0.90 0.00 0.90 0.72
X1 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00
NB1 0.96 0.91 0.86 0.99 0.63 0.63 0.60 0.90 1.00 0.83 0.68 1.00 1.00 1.00 1.00 1.00
NI1 1.00 0.62 0.78 0.99 1.00 0.95 0.82 1.00 1.00 0.35 0.92 1.00 0.99 0.70 0.70 0.98
NI5 0.77 0.31 0.15 0.36 0.93 0.57 0.44 0.57 1.00 0.68 0.37 1.00 0.63 0.09 0.00 0.00
NQ1 0.74 0.90 0.71 0.17 0.95 0.80 0.68 0.68 0.75 0.75 0.75 0.35 0.70 1.00 0.70 0.00
NQ5 0.71 0.75 0.87 0.78 1.00 0.92 1.00 1.00 0.62 0.55 0.62 0.62 0.70 0.82 0.97 0.82
NR1 0.62 0.16 0.54 0.61 0.67 0.59 0.78 0.67 0.83 0.35 0.53 0.88 0.50 0.00 0.50 0.47
NU3 0.55 0.36 0.41 0.68 0.78 0.57 0.57 0.83 0.92 0.35 0.50 0.67 0.33 0.33 0.33 0.67
NU5 0.35 0.27 0.30 0.20 0.64 0.90 0.78 0.66 0.96 0.12 0.23 0.44 0.00 0.23 0.25 0.00
NU6 0.67 0.69 0.14 0.34 1.00 0.82 0.40 0.40 0.94 0.62 0.35 1.00 0.48 0.70 0.00 0.00
NY3 1.00 0.75 0.52 0.73 1.00 0.86 0.72 0.76 1.00 0.62 0.83 0.78 1.00 0.80 0.33 0.70
HA1 0.96 0.10 0.93 0.93 1.00 1.00 1.00 1.00 0.88 0.00 0.75 0.75 1.00 0.00 1.00 1.00
HA6 0.97 0.96 0.97 0.97 1.00 1.00 1.00 1.00 0.92 0.88 0.92 0.92 1.00 1.00 1.00 1.00
HA8 0.96 0.62 0.93 0.74 1.00 1.00 1.00 1.00 0.88 0.35 0.75 0.75 1.00 0.70 1.00 0.70
HB1 0.93 0.82 0.97 0.98 0.70 0.75 0.70 0.80 0.88 0.50 1.00 1.00 1.00 1.00 1.00 1.00
HB10 1.00 0.99 0.80 0.78 1.00 0.93 1.00 0.80 1.00 1.00 0.35 0.35 1.00 1.00 1.00 1.00
HD2 0.79 0.70 0.59 0.67 0.67 0.67 1.00 1.00 1.00 0.35 0.25 0.50 0.70 0.88 0.70 0.70
HF1 0.97 0.34 0.91 0.29 0.74 0.80 0.82 0.69 1.00 0.88 0.75 0.75 1.00 0.00 1.00 0.00
HH1 0.98 0.83 0.79 0.98 1.00 1.00 1.00 1.00 1.00 0.50 0.35 1.00 0.97 0.97 0.97 0.97
HI1 1.00 0.70 0.94 1.00 1.00 0.33 1.00 1.00 1.00 0.23 0.78 1.00 1.00 1.00 1.00 1.00
HL1 1.00 0.81 0.90 0.93 1.00 1.00 0.72 1.00 1.00 0.35 0.75 0.75 1.00 1.00 1.00 1.00
IA1 0.17 0.17 0.17 1.00 0.67 0.67 0.67 1.00 0.35 0.35 0.35 1.00 0.00 0.00 0.00 1.00
IA5 0.40 0.14 0.40 0.37 1.00 0.67 1.00 0.67 1.00 0.25 1.00 1.00 0.00 0.00 0.00 0.00
IA7 0.26 0.14 0.26 0.26 0.33 0.33 0.33 0.33 0.75 0.35 0.75 0.75 0.00 0.00 0.00 0.00
IA9 0.37 0.26 0.37 0.93 0.67 0.33 0.67 0.33 1.00 0.75 1.00 1.00 0.00 0.00 0.00 1.00
IB9 0.82 0.18 0.55 0.77 1.00 0.75 0.20 1.00 1.00 0.35 0.75 0.83 0.70 0.00 0.50 0.70
IC3 0.40 0.18 0.22 0.40 1.00 0.33 0.67 1.00 1.00 0.50 0.50 1.00 0.00 0.00 0.00 0.00
ID2 0.33 0.00 0.00 0.18 0.34 0.00 0.00 0.34 1.00 0.00 0.00 0.50 0.00 0.00 0.00 0.00
IE2 0.33 0.03 0.03 0.18 0.33 0.33 0.33 0.33 1.00 0.00 0.00 0.50 0.00 0.00 0.00 0.00
IF3 0.37 0.37 0.33 0.40 0.67 0.67 0.33 1.00 1.00 1.00 1.00 1.00 0.00 0.00 0.00 0.00
IG8 1.00 1.00 0.00 0.33 1.00 1.00 0.00 0.34 1.00 1.00 0.00 1.00 1.00 1.00 0.00 0.00
0.00
0.25
0.50
0.75
1.00

a **HA8: the reference state changes the barrier**



b **HH1: the same barrier, different evidence closure**



c **NU6: an energy difference needs a pathway identity**

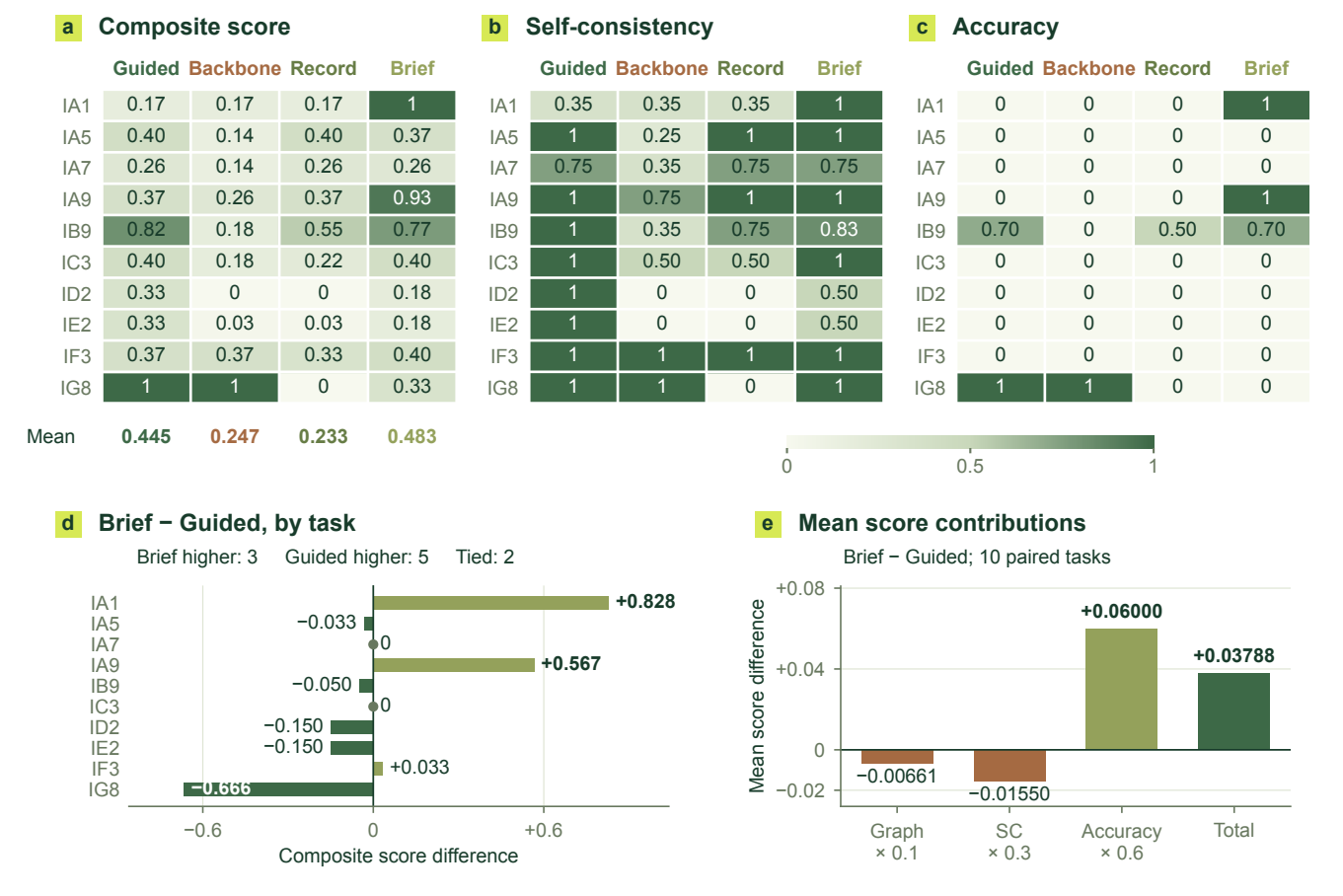
Concerted TS −247.030042392 Ha

Betaine INT −247.060060699 Ha

Difference = 18.836772 kcal mol^{-1}

Guided: concerted saddle region;
target INT → P saddle unresolved

Backbone: step-2 closure = 18.837


a Composite score
Guided Backbone Record Brief
IA1 0.17 0.17 0.17 1
IA5 0.40 0.14 0.40 0.37
IA7 0.26 0.14 0.26 0.26
IA9 0.37 0.26 0.37 0.93
IB9 0.82 0.18 0.55 0.77
IC3 0.40 0.18 0.22 0.40
ID2 0.33 0 0 0.18
IE2 0.33 0.03 0.03 0.18
IF3 0.37 0.37 0.33 0.40
IG8 1 1 0 0.33
Mean 0.445 0.247 0.233 0.483
b Self-consistency
Guided Backbone Record Brief
IA1 0.35 0.35 0.35 1
IA5 1 0.25 1 1
IA7 0.75 0.35 0.75 0.75
IA9 1 0.75 1 1
IB9 1 0.35 0.75 0.83
IC3 1 0.50 0.50 1
ID2 1 0 0 0.50
IE2 1 0 0 0.50
IF3 1 1 1 1
IG8 1 1 0 1
c Accuracy
Guided Backbone Record Brief
IA1 0 0 0 1
IA5 0 0 0 0
IA7 0 0 0 0
IA9 0 0 0 1
IB9 0.70 0 0.50 0.70
IC3 0 0 0 0
ID2 0 0 0 0
IE2 0 0 0 0
IF3 0 0 0 0
IG8 1 1 0 0
0 0.5 1
d Brief − Guided, by task
Brief higher: 3 Guided higher: 5 Tied: 2
IA1 +0.828
IA5 −0.033
IA7 0
IA9 +0.567
IB9 −0.050
IC3 0
ID2 −0.150
IE2 −0.150
IF3 +0.033
IG8 −0.666
−0.6 0 +0.6
Composite score difference
e Mean score contributions
Brief − Guided; 10 paired tasks
Mean score difference
+0.08 +0.04 0 −0.02
−0.00661
−0.01550
+0.06000
+0.03788
Graph × 0.1
SC × 0.3
Accuracy × 0.6
Total